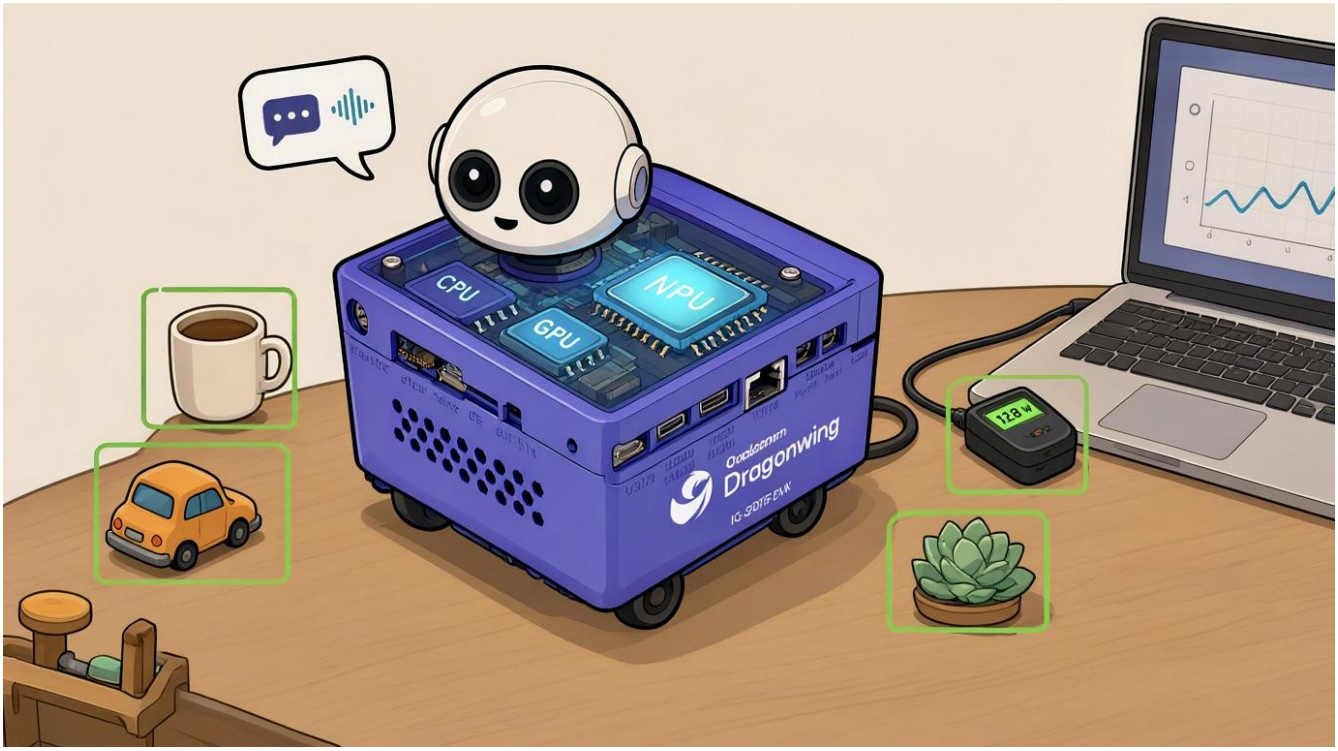
Qualcomm Dragonwing IQ-9075 EVK: first tests



Cover image generated locally with Qwen-Image 2.1 under the author's direction.

Oct 14, 2026, updated Oct 26, 2026 with energy measurements and Oct 27–28, 2026 with Jetson Orin Nano Super and Raspberry Pi 5 comparisons · @Marcelo Rovai

In one afternoon, without flashing a new OS, the IQ-9075 EVK showed what a heterogeneous SoC delivers: **the NPU sped up INT8 CNNs by 11 to 95 times, ran object detection at 30 FPS with headroom to spare, and did the prefill of a 4-billion-parameter LLM 18 times faster than the CPU**. Decode, on the other hand, stayed bound by memory bandwidth, whichever processor ran it. A second session, measuring power at the board's input, showed that the **NPU's advantage is even larger in energy: 5 to 70 times less per inference on the vision models, and two-thirds of the CPU's energy per token on LLM decode, at the same speed**. Next to a Jetson Orin Nano Super running the same LLM files, the Jetson's GPU decoded 15 to 20% faster, but on the **4B model the IQ-9075's NPU did the prefill faster and used less energy in both phases**. Without an NPU bundle, as with the newer Qwen3.5-4B, the Jetson won on both counts, and a Raspberry Pi 5 with the same llama.cpp decoded 4 to 5 times slower than the IQ-9075's CPU.

> We hope Qualcomm keep updating the LLM models compatibility with the NPU (at least the most capable ones).

The kit

The IQ-9075 EVK is Qualcomm's evaluation board for the Dragonwing IQ-9075 SoC, part of the **IQ9 series, aimed at industrial IoT, robotics and Edge AI in demanding environments**. The SoC sits on a module (IQ-9075M) with four PMICs and three LPDDR5 chips, mounted on a mainboard with connectors and slots for mezzanine boards.

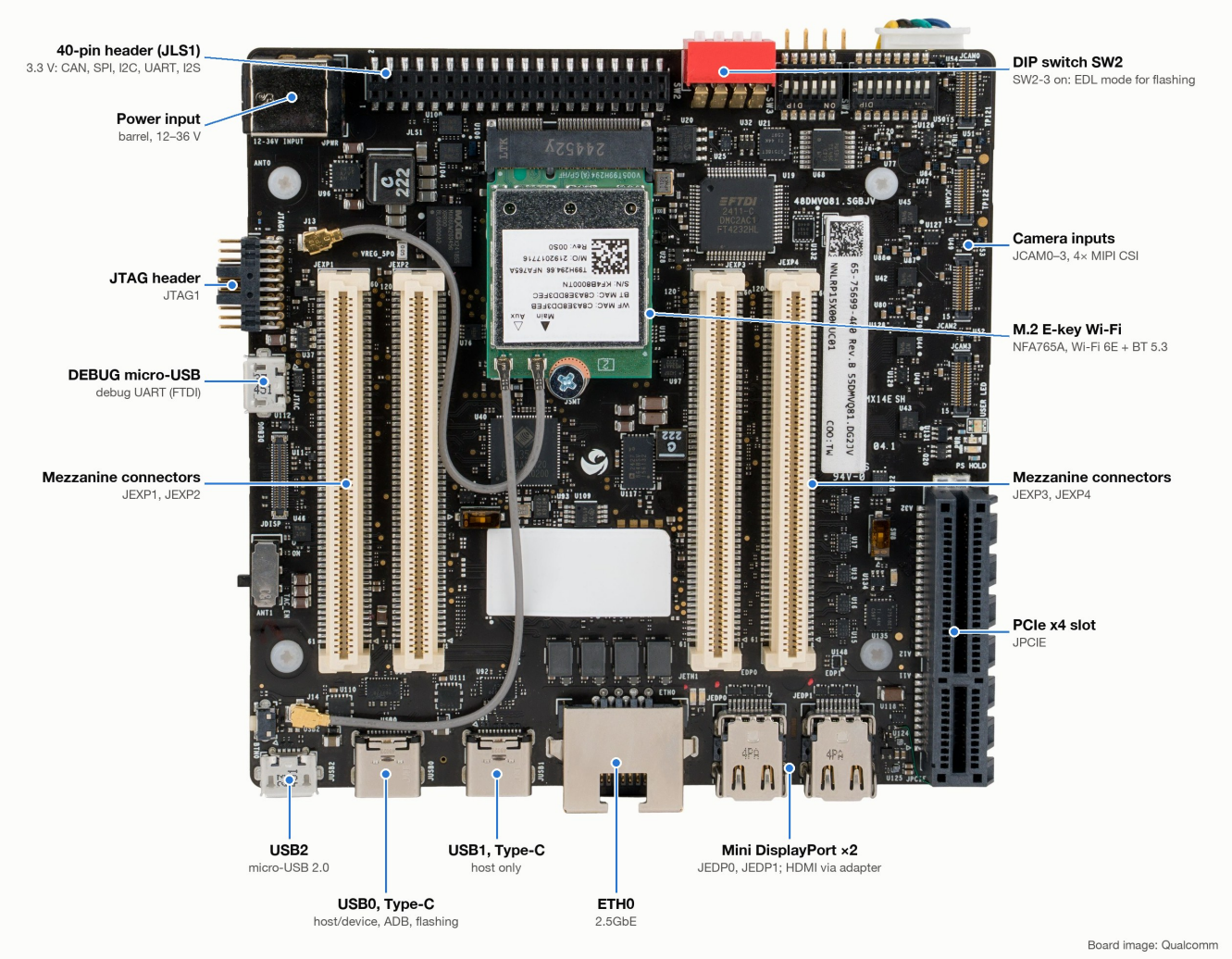


Image: Qualcomm.

Block	Specification
CPU	8× Kryo Gen 6 cores, up to 2.36 GHz
GPU	Adreno 663
NPU	2 Hexagon Tensor Processors (HVX + HMX), 100 dense / 200 sparse TOPS (INT8)
Real time and safety	4-core MCU with its own I/Os, plus an isolated Safety Island (SAIL)
Memory	36 GB LPDDR5 at 3200 MHz (96-bit bus), with inline ECC
Storage	UFS, 107 GB visible to the OS
Video	decode of 1× 8K60 or 4× 4K60 (AV1, H.264, H.265, VP9)
Cameras	4 MIPI CSI ports, plus USB and Ethernet cameras
Connectivity	Wi-Fi 6E and Bluetooth 5.3 (NFA765A module), 2.5GbE, CAN-FD
Power	2.10 mm barrel jack, 12 V to 36 V
Software	Qualcomm Linux (Yocto) or Ubuntu 24.04, with 10+ years of longevity support

Two details deserve attention. The IQ-9075 comes in two variants, 50 TOPS (two PMICs) and 100 TOPS (four PMICs), and the EVK uses the 100 TOPS one. And theoretical memory bandwidth, estimated from the 96-bit bus at 6400 MT/s, is around 77 GB/s. That number, not the TOPS, explains most of the LLM results.

The mainboard concentrates the connectors used in this article: the barrel power input, the SW2 DIP switch that puts the board in flashing mode, the USB0 port used for flashing, the mini DisplayPorts for the monitor, and the camera inputs.



Board image: Qualcomm; labels added for this article.

Where it fits

The IQ-9075 doesn't compete on raw power with NVIDIA's top line; its edge is a dense NPU, ECC memory, a real-time subsystem, and an industrial lifecycle. The table uses reference values from vendor spec sheets; check them before citing.

	Arduino UNO Q	Arduino VENTUNO Q	Raspberry Pi 5	Jetson Orin Nano Super	Jetson AGX Orin 64GB	IQ-9075 EVK
Class	SBC + MCU, real projects, prototyping, and education	SBC + MCU, AI and robotics	general-purpose SBC	entry Edge AI	high-end Edge AI	industrial Edge AI
CPU	4× A53 at 2.0 GHz (QRB2210) + STM32U585	8× Kryo Gen 6 (IQ-8275): 4 performance cores at up to 2.35 GHz + 4 efficiency cores, + STM32H5F5	4× A76 at 2.4 GHz	6× A78AE at 1.7 GHz	12× A78AE at 2.2 GHz	8× Kryo Gen 6 (QCS9075): 8 performance cores (A78C) at 2.36 GHz

	Arduino UNO Q	Arduino VENTUNO Q	Raspberry Pi 5	Jetson Orin Nano Super	Jetson AGX Orin 64GB	IQ-9075 EVK
AI accelerator	no dedicated NPU (Hexagon DSP for sensors and audio)	Hexagon NPU	none on the SoC; add-ons: Hailo (AI HAT+, AI HAT+ 2), MemryX MX3 (M.2)	Ampere GPU	Ampere GPU + DLA	Hexagon NPU
Stated AI (INT8)	—	40 dense TOPS	13–26 TOPS (AI HAT+), 40 TOPS INT4 (AI HAT+ 2), 24 TOPS (MX3)	~67 sparse TOPS	~275 sparse TOPS	100 dense / 200 sparse
RAM	2–4 GB	16 GB	up to 16 GB	8 GB	64 GB	36 GB with ECC
Real-time	yes (MCU)	yes (MCU)	no	no	no	yes (MCU + SAIL)
AI ecosystem	App Lab, Edge Impulse, Debian (llama.cpp on the CPU)	App Lab, Ubuntu/Debian	Raspberry Pi OS (Debian), broad	CUDA, TensorRT	CUDA, TensorRT	QNN/QAIRT, LiteRT, AI Hub, GenieX, Edge Impulse

Two caveats. NVIDIA quotes sparse TOPS while Qualcomm highlights dense ones; on a sparse basis, the IQ-9075 sits between the Orin Nano and the AGX Orin. And for LLMs, what matters is memory bandwidth: about 77 GB/s on the IQ-9075 versus about 205 GB/s on the AGX Orin.

There is also a continuity across the Qualcomm boards: the Arduino UNO Q uses an entry-level Dragonwing (QRB2210), and the Arduino VENTUNO Q steps up to a Dragonwing IQ8 (IQ-8275), a smaller sibling of the IQ-9075 with half the performance cores (4 against 8), 40 TOPS instead of 100, and 16 GB of RAM. The same software ecosystem spans from the maker board to the industrial one, and what students learn on one carries over to the others.

First contact: the factory image

The board arrived with Qualcomm Linux 1.6 already flashed, not empty as some documentation pages suggest. It is Qualcomm's Yocto distribution, with a Weston/Wayland desktop and a complete set of AI tools. Connected to a monitor (HDMI through an adapter), a keyboard and Wi-Fi, it opens a graphical terminal as root.

SSH access

Typing on the board's terminal is slow and does not allow pasting, so the first useful step is SSH. The server is already running and Qualcomm Linux's default root password is `oelinux123`:

1. On the board, `ip -4 a` shows the IP of the `wlan0` interface.
2. On the host, `ssh root@<IP>` with the default password.
3. Change the password with `passwd` on first login.

What the image includes

Item	Result
OS	Qualcomm Linux 1.6-ver.1.2.1, kernel 6.6.97
CPU and memory	8 cores; 34 GiB visible, 32 free at idle
Storage	107 GB UFS, 99 GB free

Item	Result
File system	OSTree: `/usr` read-only; `/`, `/var`, `/root` and `/usr/local` writable
Multimedia	GStreamer 1.22 with the IM SDK plugins (`qtimltflite`, `qtimlqnn`, `qtimlsnpe`, detection, pose, segmentation)
AI runtimes	QNN (CPU, DSP, HTP V73), SNPE, TFLite with QNN delegate, GenAI Transformer libraries
Tools	`qnn-net-run`, `snpe-net-run`, `benchmark_model`, `python3` 3.12, `docker` 25
Ready-made models	10 TFLite and 3 DLC models in `/etc/models`, with configs in `/etc/configs`
Missing	`apt`, `pip`, a compiler, `v4l2-ctl`
Idle temperature	42–45 °C on every zone (CPU, NPU, DDR)

The lack of `apt` and a compiler sets the strategy: anything not in the image comes in through Docker. The bundled models and sample apps, on the other hand, make it possible to measure the NPU without downloading anything.

Vision: CPU, GPU and NPU

The NPU sped up INT8 convolutional models by 11 to 95 times, but lost to the CPU on the one model dominated by data movement. All tests used TFLite's `benchmark_model` with the models in `/etc/models`, batch 1: CPU with 8 threads (XNNPACK) and NPU through the QNN delegate on the HTP backend, in high-performance mode.

One model, three processors

On the quantized Inception v3, the whole graph was delegated to the NPU.

Backend	Mean latency	Throughput	vs CPU
CPU, 8 threads	15.4 ms (range 11.7 to 51 ms)	~65 FPS	1×
Adreno 663 GPU	62 ms	~16 FPS	4× slower
Hexagon NPU (HTP V73)	0.76 ms (std. dev. 15 µs)	~1,300 FPS	~20× faster

The GPU lost because TFLite's GPU delegate works in FP16/FP32 and dequantizes the INT8 model. Beyond speed, the NPU was far more stable, which matters in real-time applications. At about 5.7 GMACs per inference, 0.76 ms works out to roughly 15 effective TOPS, well below the 100 TOPS peak: batch 1 and a small model don't fill the accelerator.

The ten models in the image

Model	CPU (ms)	NPU (ms)	Speedup
face_attrib_net_quantized	54.15	0.568	95.4×
inception_v3_quantized	13.98	0.737	19.0×
hrnet_pose_quantized	19.79	1.261	15.7×
midas_quantized	19.68	1.250	15.7×
deeplabv3_plus_mobilenet_quantized	62.91	5.263	12.0×
yolox_quantized	36.06	3.157	11.4×
face_det_lite_quantized	4.75	0.558	8.5×

Model	CPU (ms)	NPU (ms)	Speedup
facemap_3dmm_quantized	0.82	0.230	3.6×
yamnet (float)	0.90	0.272	3.3×
quicksrnetsmall_quantized	3.08	3.786	0.8×

Three patterns emerge:

- **Compute-heavy INT8 CNNs gain 11 to 95 times.** This is the NPU's sweet spot. The extreme gain on face_attrib suggests the model uses INT8 operators that XNNPACK handles poorly.
- **Tiny models gain only 3 to 8 times.** The NPU hits a floor of 0.2 to 0.3 ms, the fixed cost of dispatching work to the DSP over FastRPC. yamnet is float and still sped up, because the HTP V73 runs FP16.
- **Super-resolution loses on the NPU.** quicksrnet does little math and moves large input and output images; the time goes to transfers, not compute.

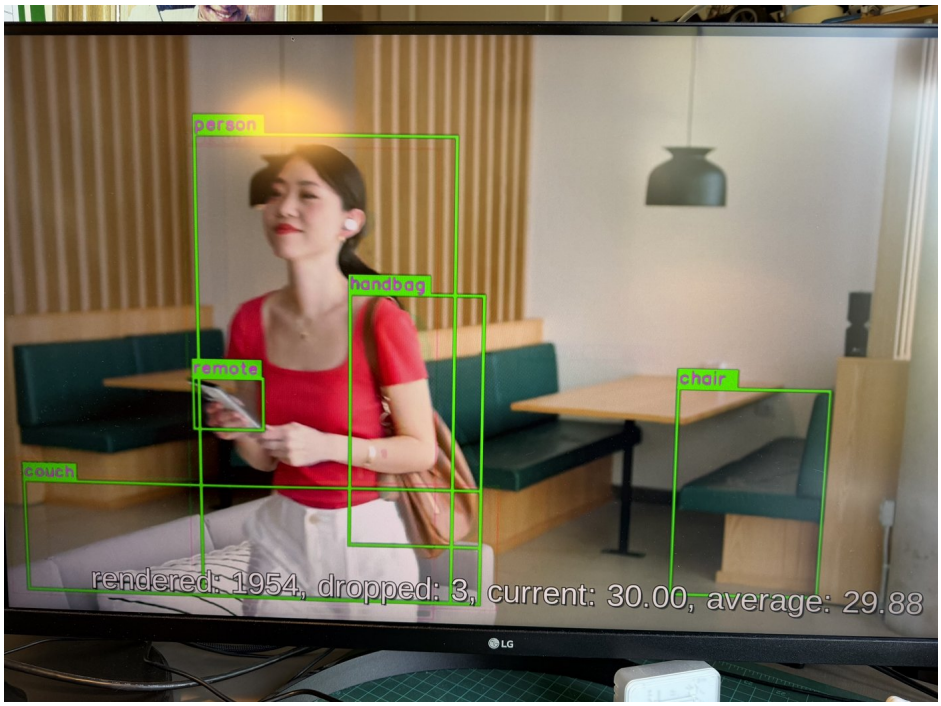
The NPU's gain depends on the model's arithmetic intensity, that is, how much compute it does per byte moved, far more than on peak TOPS.

Video pipeline: real-time object detection

YOLOX ran at a steady 30 FPS over a 1 min 41 s video, dropping 3 frames out of nearly 2,000, and the limit was the video's own frame rate, not the board. Since YOLOX takes about 3 ms on the NPU, there is roughly ten times headroom for more streams.

The `gst-ai-object-detection` app builds the whole pipeline from a JSON file: hardware H.264 decode, preprocessing, TFLite inference on the NPU (`"runtime": "dsp"`), postprocessing and display through `waylandsink`. To run it over SSH and show it on the board's monitor, point it at the Weston display:

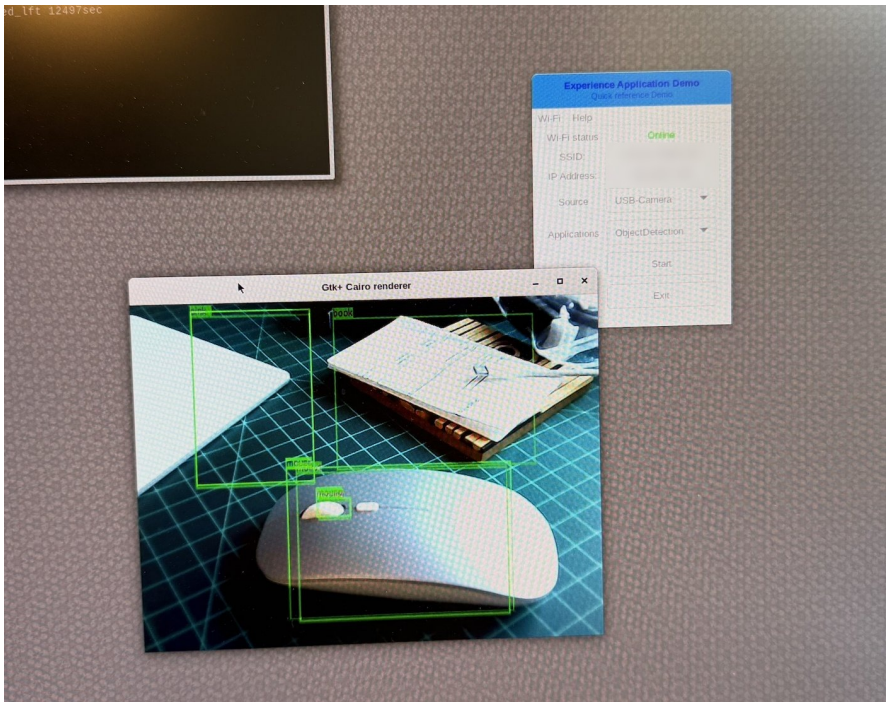
```
export XDG_RUNTIME_DIR=/dev/socket/weston
export WAYLAND_DISPLAY=wayland-1
gst-ai-object-detection --config-file=/etc/configs/config_detection.json
```



The logs are noisy and alarming the first time. The HTP compiler warnings (`conv_tile_cost`, `concat`), the GBM `mmap` failures and `Failed to link pad` (the MP4's audio track) are harmless. The useful line is the HTP bandwidth summary: `spill_bytes=0` means the model fit entirely in the NPU's local memory.

USB camera and Qdemo

The USB webcam showed up as `/dev/video0`, but the factory config (YUY2 at 1920×1080, 30 FPS) failed with `not-negotiated`: at high resolutions, most webcams only deliver 30 FPS in MJPEG. The simplest path was Qdemo ("Experience Application Demo"), Qualcomm Linux's graphical menu that builds the same pipelines with a choice of source and application.



The model's mistakes make good classroom examples. A phone became "remote", pendant lamps became "vase," and a laptop screen became "tv": a COCO-trained detector confuses objects that look like classes it knows. On the live camera, the mouse got two overlapping boxes, a non-maximum suppression (NMS) case that failed to remove the redundant detection at a 40% threshold.

Local LLMs with GenieX

On a 4-billion-parameter model, the NPU did the prefill 18 times faster than the CPU, but decode was practically the same on both, around 16 to 18 tokens/s. Decode is bound by memory bandwidth; prefill, by compute. The same speed did not mean the same energy, as the energy section shows.

How to run it

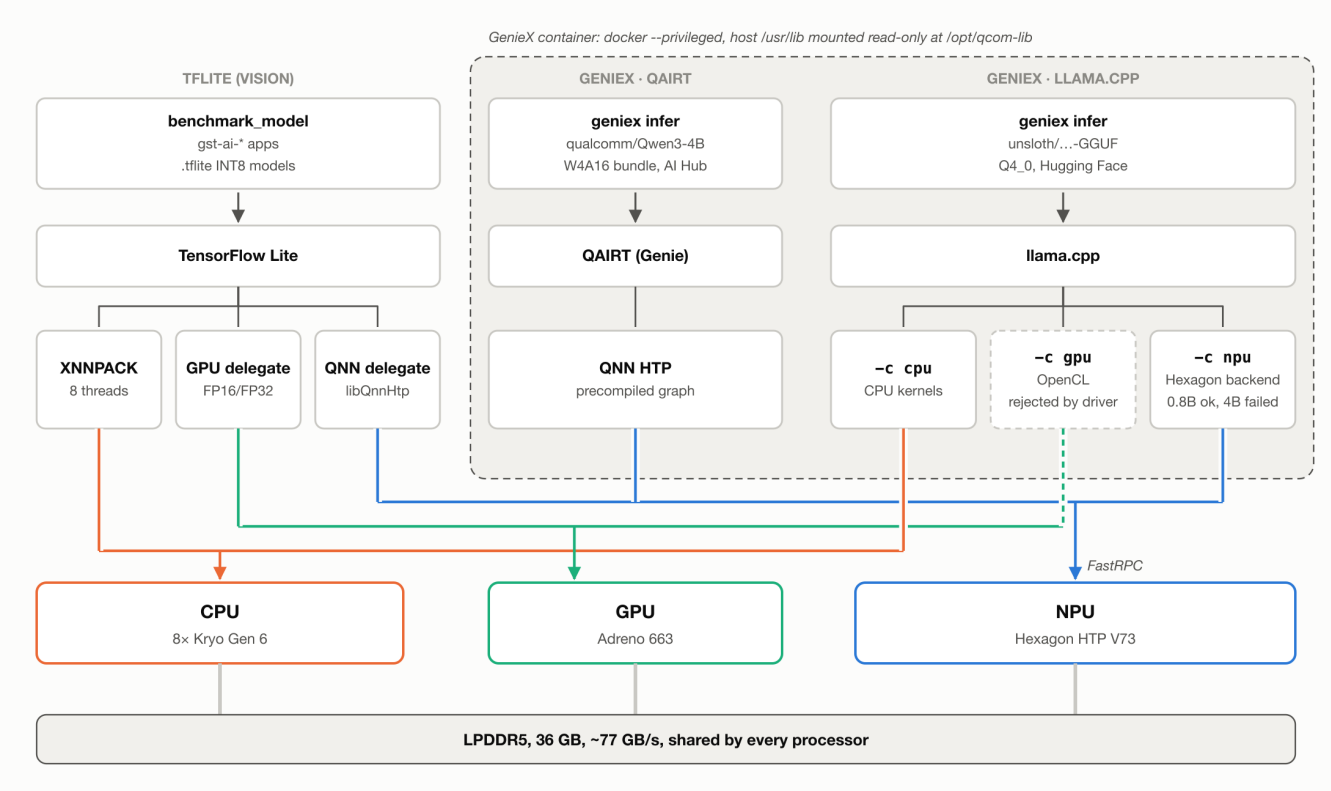
With no `apt` and no compiler, the path was GenieX, which Qualcomm publishes as an ARM64 container for the QCS9075. It offers two runtimes: `llama_cpp`, for GGUF models from Hugging Face, and `qairt`, for precompiled bundles from Qualcomm AI Hub. The container reaches the NPU through `--privileged` and by mounting the host's libraries:

```
mkdir -p /root/geniex-data
docker pull docker.io/qualcomm/geniex:latest
docker run -it --rm --privileged \
  -e GENIEX_DATADIR=/data \
  -v /root/geniex-data:/data \
  -v /usr/lib:/opt/qcom-lib:ro \
  docker.io/qualcomm/geniex:latest
```

`GENIEX_DATADIR` is essential: without it, models live inside the container and vanish on every `--rm`. For benchmarks, turn off thinking mode (`--think=false`) and redirect input so the command exits on its own:

```
geniex infer <model> -c npu|cpu --think=false --max-tokens 256 -p "<prompt>" < /dev/null
```

The diagram below summarizes the three software paths used in these tests, from the command to the processor.



Results

The short prompt was “Explain in three paragraphs what TinyML is”, in Portuguese. The long one had 1,087 Portuguese words: 1,882 tokens for Qwen3 and 1,641 for Qwen3.5. Time to first token measures prefill.

Model	Runtime	Prompt	Decode (tok/s)	First token
Qwen3.5-0.8B Q4_0	llama.cpp, NPU	short	29.7	0.1 s
Qwen3.5-0.8B Q4_0	llama.cpp, CPU	short	57.1	0.1 s
Qwen3.5-0.8B Q4_0	llama.cpp, NPU	long	28.4	1.7 s
Qwen3.5-0.8B Q4_0	llama.cpp, CPU	long	49.4	5.1 s
Qwen3-4B Q4_0	llama.cpp, NPU	short	failed (`SDKError`)	—
Qwen3-4B Q4_0	llama.cpp, CPU	short	17.9	0.4 s
Qwen3-4B W4A16	QAIRT, NPU	short	17.9	0.1 s
Qwen3-4B Q4_0	llama.cpp, CPU	long	13.9	31.1 s
Qwen3-4B W4A16	QAIRT, NPU	long	16.1	1.7 s

The GPU could not be tested: the shader compiler in the host’s Adreno driver (qcom-adreno 1.808.11) rejected llama.cpp’s OpenCL kernels.

What the numbers say

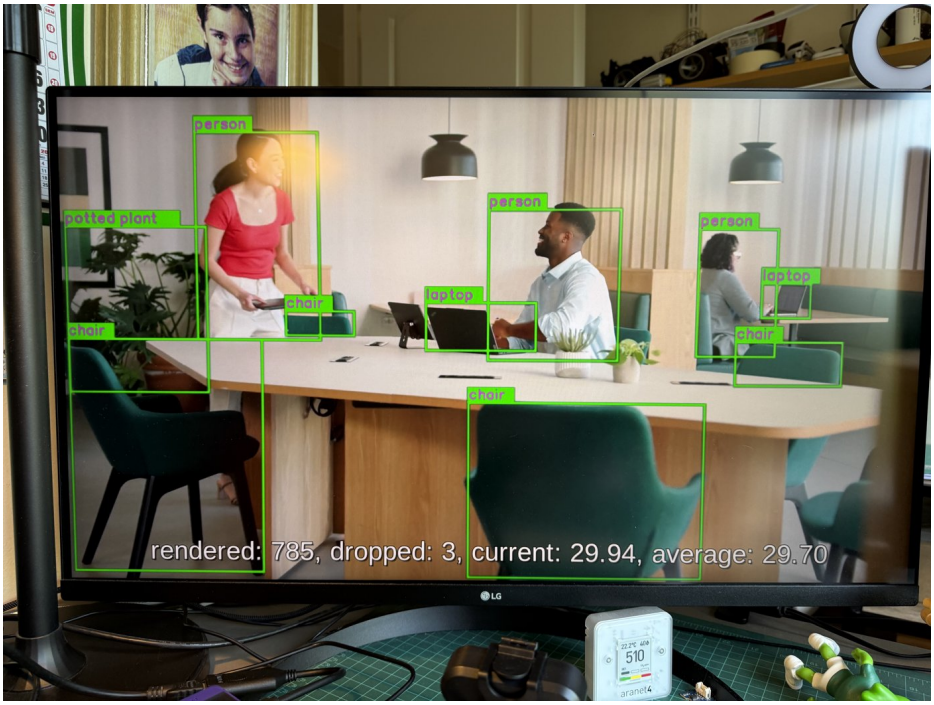
- **Decode is memory-bound.** On the 4B model, CPU and NPU both gave 17.9 tok/s. With about 2.3 GB of weights read per token, that is roughly 41 GB/s effective, 55% of the theoretical 77 GB/s. More TOPS do not speed up decode.
- **Prefill is compute-bound.** With the long prompt, the 4B model took 31.1 s to the first token on the CPU and 1.7 s on the NPU: about 60 versus more than 1,000 tokens/s processed. For RAG, agents or document summarization, that is the difference between a usable and an unusable application.

- **Long context also costs at decode.** With about 1,900 tokens in the KV cache, decode dropped 22% on the CPU and 10% on the NPU, because each new token reads the weights plus the whole cache.
- **On small models, the CPU wins.** On the 0.8B model, the CPU generated nearly twice the tokens/s of the NPU. With little math per token, the fixed cost of dispatching each layer to the DSP dominates.
- **The right NPU runtime is QAIRT.** llama.cpp's Hexagon backend lost to the CPU on the 0.8B model and failed on the 4B, probably due to a FastRPC session limit with the Qualcomm Linux 1.6 driver. The AI Hub bundle ran without issues.

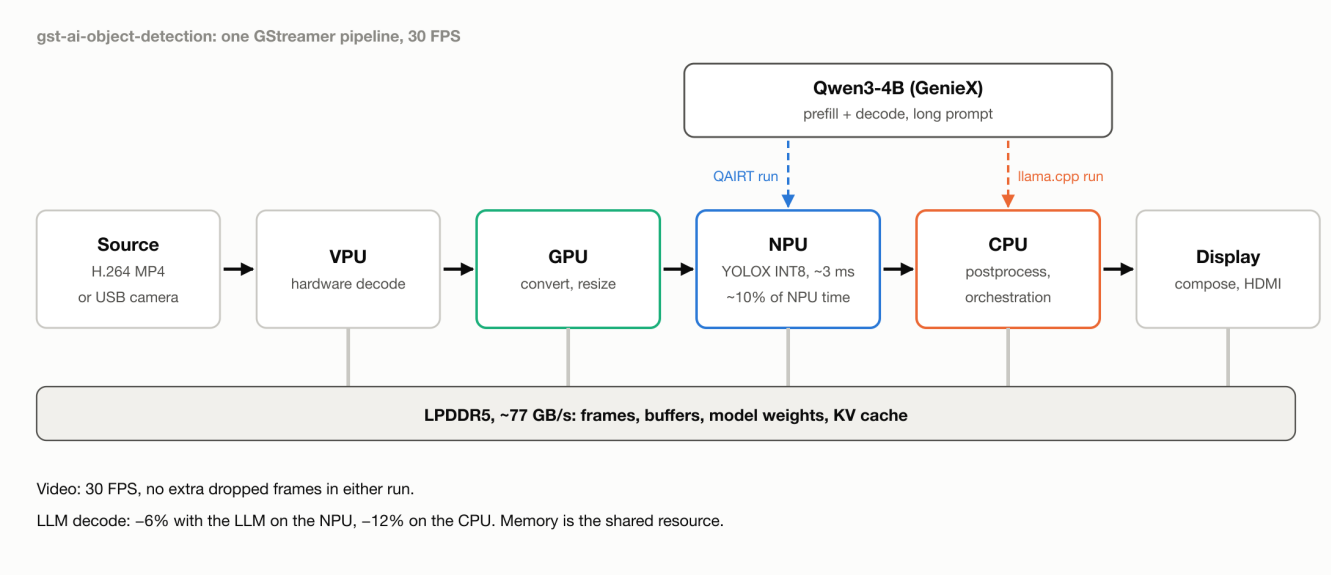
Concurrent load: LLM and vision together

With YOLOX processing video at 30 FPS, neither the CPU nor the NPU LLM made the video drop a single frame; the cost showed up only in the LLM, as a 6% to 12% decode slowdown. The test used two SSH sessions: one running `gst-ai-object-detection` on the monitor and another running GenieX with Qwen3-4B on the long prompt.

LLM (Qwen3-4B)	Prefill alone	Prefill with video	Decode alone	Decode with video	Video
NPU (QAIRT)	1.7 s	1.7 s	16.1 tok/s	15.2 tok/s (–6%)	30 FPS, no extra dropped frames
CPU (llama.cpp)	31.1 s	33.0 s (+6%)	13.9 tok/s	12.3 tok/s (–12%)	30 FPS, no extra dropped frames



The video did not suffer even with all 8 cores saturated because the vision pipeline barely uses the CPU: decode runs on the VPU, scaling on the GPU, YOLOX on the NPU and composition on the display processor. The CPU only orchestrates. With the LLM also on the NPU, both workloads shared the accelerator with no visible conflict; YOLOX takes about 10% of the NPU's time at 30 FPS.



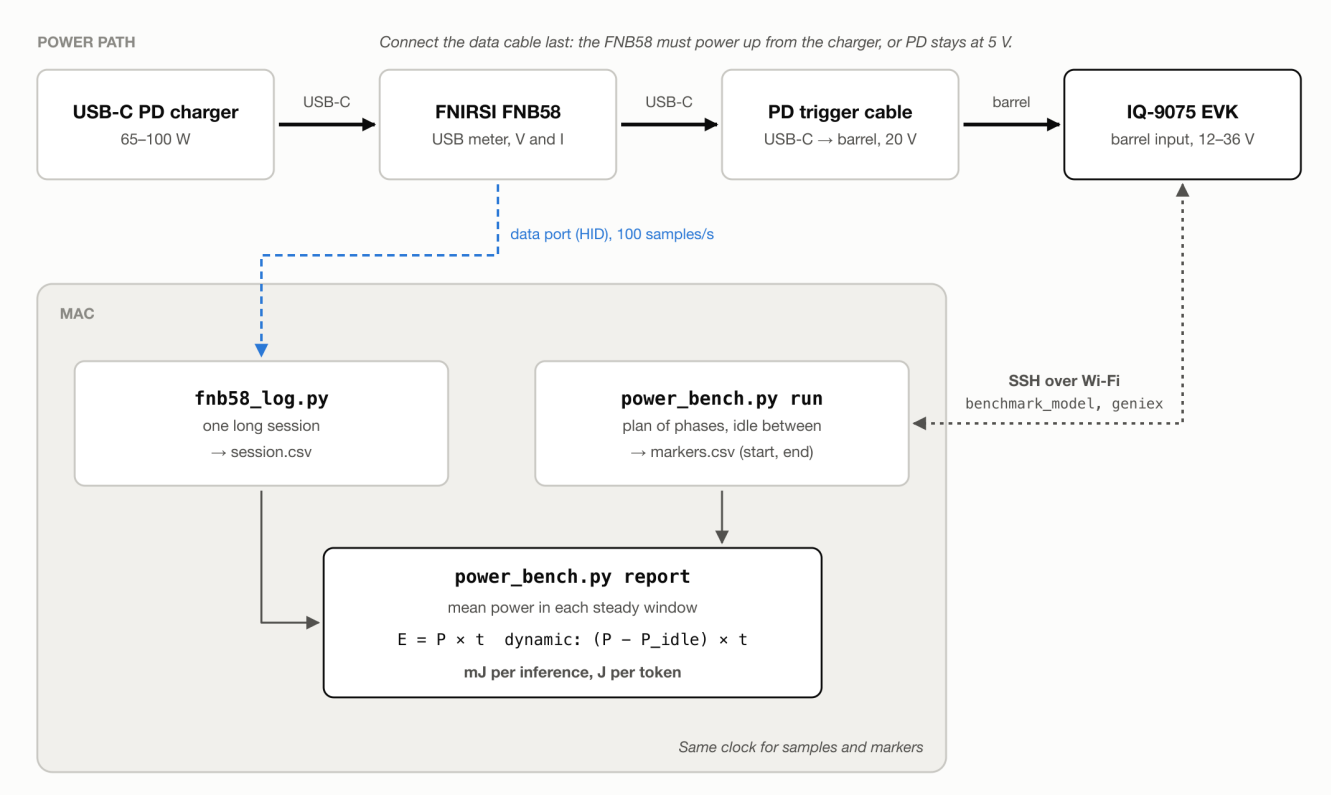
The resource everyone competes for is memory. Decoded video, buffers and inference all read the same LPDDR5, and LLM decode, already bandwidth-bound, is the first to feel it. The CPU lost more, probably because it also competes for cache and scheduling with the GStreamer threads.

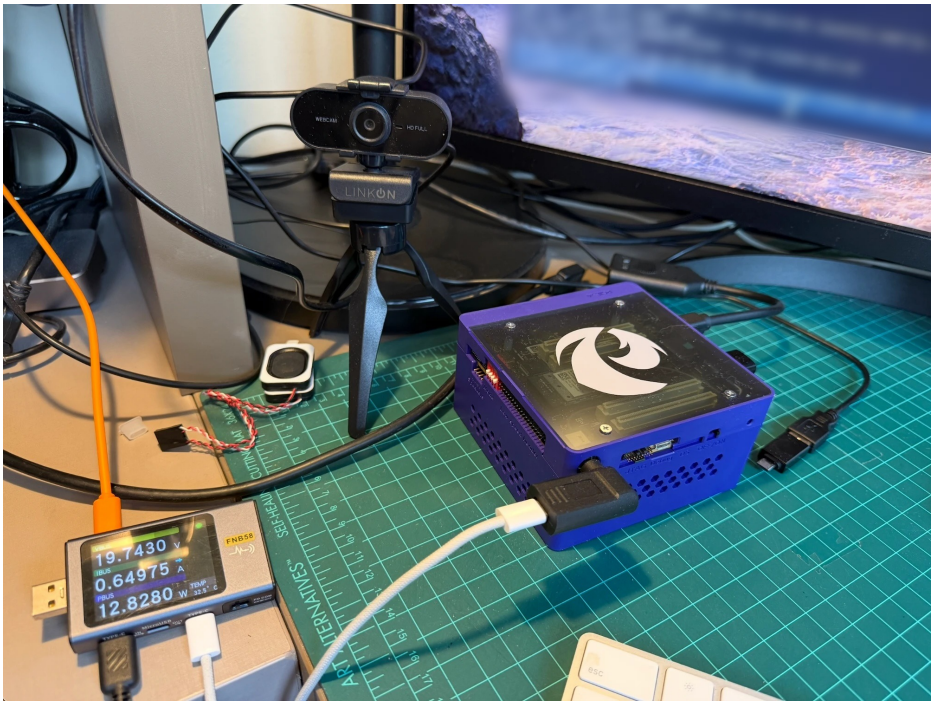
Energy: what each inference costs

Measured at the board's power input, the NPU used 5 to 70 times less energy per inference than the CPU on the vision models, and 20 times less on the prefill of a 4B LLM. Even where it was not faster, it rarely cost more: on 4B decode it matched the CPU's speed with two-thirds of the energy, and on super-resolution, 18% slower, it still used 9% less. The exception was the 0.8B LLM on llama.cpp's Hexagon backend, which lost on both speed and energy.

How it was measured

The board ran from a USB-C PD charger through a FNIRSI FNB58 USB meter and a PD trigger cable fixed at 20 V, ending in the barrel jack. The FNB58 sent 100 samples per second to a Mac over its data port, while a script launched each workload over SSH and marked its start and end on the same clock. Energy per inference is the average power over the steady part of each run multiplied by the mean latency. "Total" is everything the board drew; "dynamic" subtracts the idle power measured just before each run.





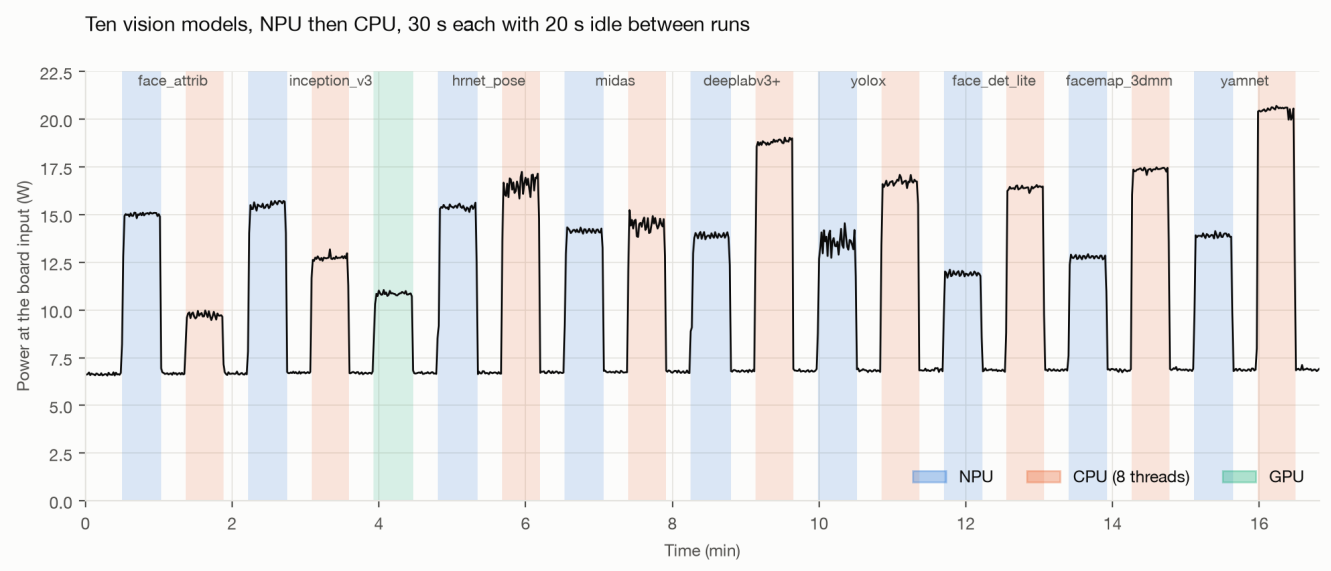
Configuration	Idle power
Monitor (HDMI), Bluetooth keyboard and mouse, USB webcam, Wi-Fi	~8.2 W
Wi-Fi only (all measurements below)	6.6 to 6.9 W, rising slowly as the board warmed

Two practical notes. The order of connection matters: the trigger cable only negotiated 20 V when the FNB58 powered up from the charger. With the meter already powered by the Mac, the board got stuck at 5 V and did not boot (green LED blinking fast). And on macOS the meter accepted only one logging session per power cycle, so each battery of tests ran under a single long session.

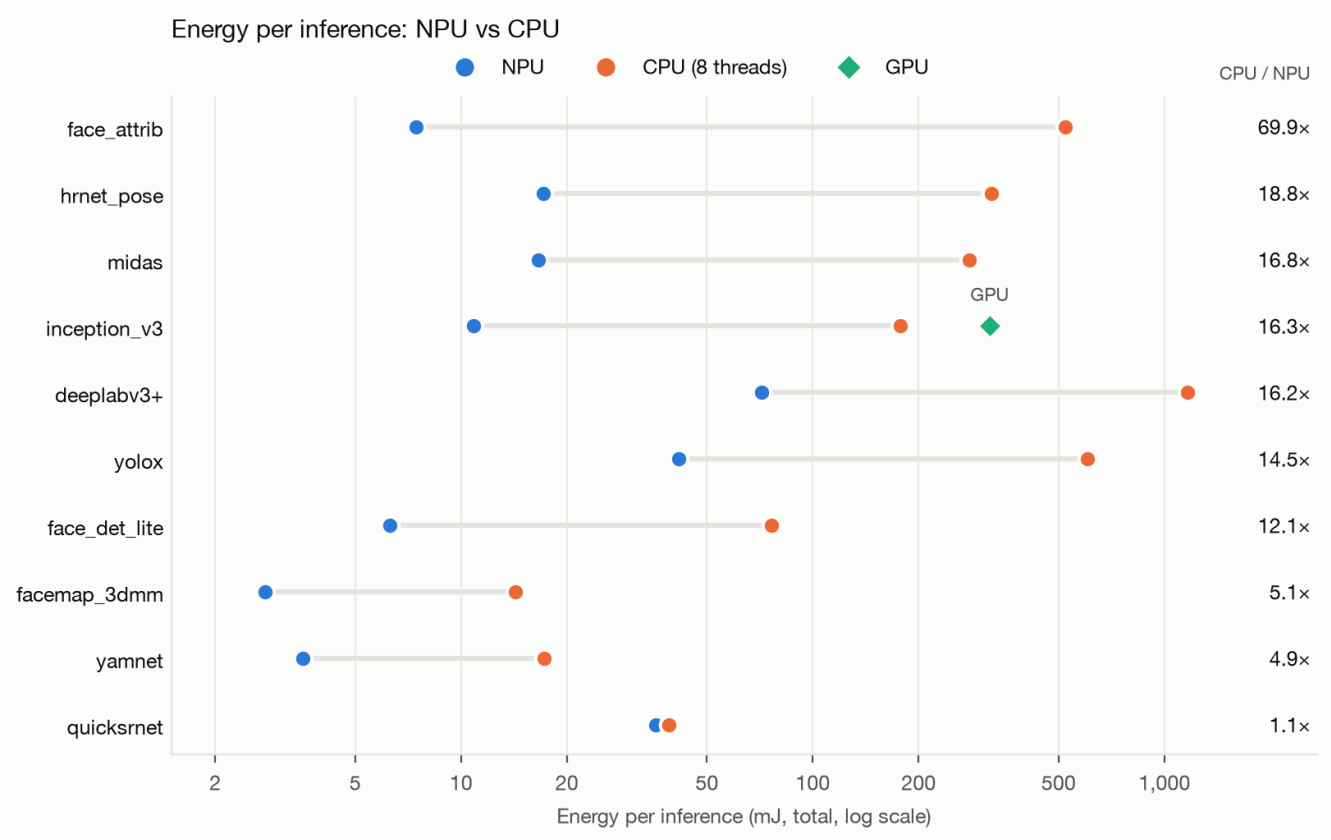
The numbers cover the whole board, not just the SoC: memory, Wi-Fi, fan, and power conversion all count. The comparison between processors holds because everything else stays the same. Repeating Inception v3 on the NPU in a second session gave 10.89 mJ against 10.87 mJ in the first.

Vision models

Same ten models and settings as the vision tests, with 30 s runs and 20 s of idle between them. Energy is total energy per inference.



Model	Speedup (latency)	NPU power	CPU power	NPU energy	CPU energy	Energy ratio
face_attrib_net	108×	15.0 W	9.7 W	7.5 mJ	521 mJ	70×
inception_v3	19.9×	15.5 W	12.7 W	10.9 mJ	177 mJ	16.3×
hrnet_pose	17.4×	15.4 W	16.6 W	17.1 mJ	321 mJ	18.8×
midas	16.4×	14.1 W	14.5 W	16.6 mJ	278 mJ	16.8×
deeplabv3_plus_mobilenet	12.0×	13.9 W	18.8 W	71.6 mJ	1,164 mJ	16.2×
yolox	11.8×	13.5 W	16.8 W	41.5 mJ	604 mJ	14.5×
face_det_lite	8.8×	11.9 W	16.4 W	6.3 mJ	76 mJ	12.1×
facemap_3dmm	3.8×	12.8 W	17.4 W	2.8 mJ	14.3 mJ	5.1×
yamnet (float)	3.3×	13.9 W	20.5 W	3.6 mJ	17.2 mJ	4.9×
quicksrnetsmall	0.85×	9.7 W	12.5 W	35.8 mJ	39.1 mJ	1.09×



The GPU, tested on Inception v3 only, drew 10.9 W and used 318 mJ per inference: 1.8 times the CPU and 29 times the NPU. Its latency in this run was 29 ms, against 62 ms on October 14; the difference is not explained yet.

- **Power is not energy.** On Inception v3 the NPU drew more power than the CPU (15.5 W against 12.7 W), but it finished each inference 20 times sooner.
- **On most models the energy gain exceeds the speedup.** The NPU drew 12 to 15.5 W, while 8 CPU threads drew 16 to 20.5 W, so the NPU was both faster and lower-power.
- **Tiny models still save about 5 times.** The FastRPC dispatch floor costs time, but little energy.
- **Slower on the NPU did not mean more expensive.** quicksrnet ran 18% slower on the NPU, but at 9.7 W against 12.5 W, so it still used 9% less total energy and 1.65 times less dynamic energy.
- **face_attrib explains its own speedup.** On the CPU it drew only 9.7 W, 3 W above idle, while the other CPU runs drew 16 to 20 W. The cores were mostly waiting, which supports the guess that XNNPACK runs some of its INT8 operators on slow, poorly parallelized kernels.

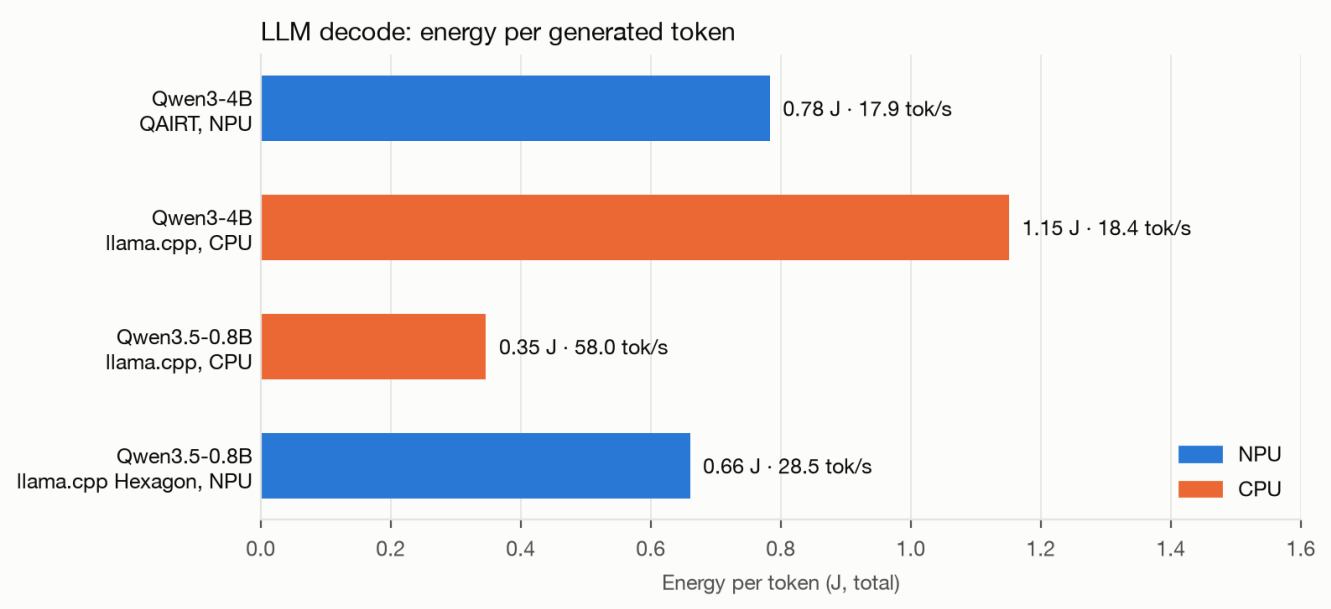
LLMs

Same prompts and models as the LLM tests. The prefill and decode windows were cut from the power trace using the end of generation, stamped inside the container, plus the time to first token and decode rate reported by GenieX.

Decode, short prompt, 256 tokens:

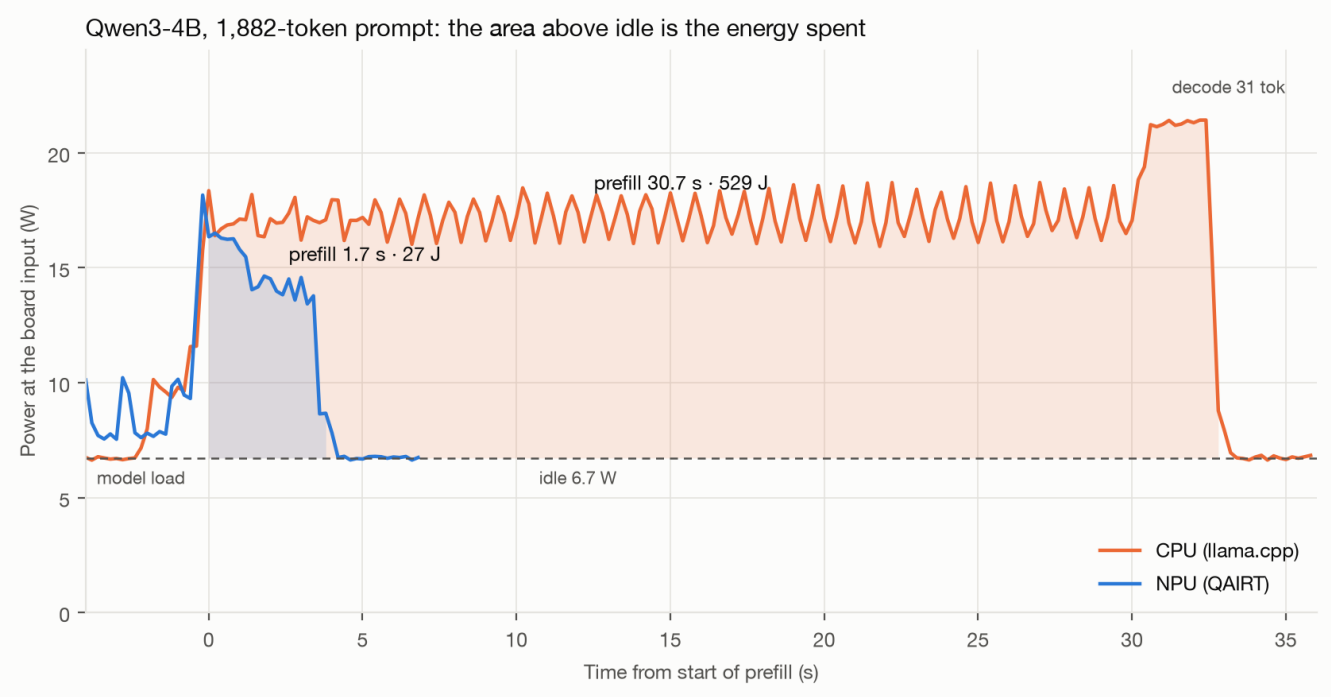
Model	Runtime, processor	tok/s	Power	Energy per token	Tokens per Wh
Qwen3-4B W4A16	QAIRT, NPU	17.9	14.0 W	0.78 J	~4,600
Qwen3-4B Q4_0	llama.cpp, CPU	18.4	21.2 W	1.15 J	~3,100
Qwen3.5-0.8B Q4_0	llama.cpp, CPU	58.0	20.1 W	0.35 J	~10,400

Model	Runtime, processor	tok/s	Power	Energy per token	Tokens per Wh
Qwen3.5-0.8B Q4_0	llama.cpp (Hexagon), NPU	28.5	18.8 W	0.66 J	~5,400



Prefill, long prompt (1,882 tokens for Qwen3, 1,641 for Qwen3.5):

Model	Runtime, processor	Time to first token	Power	Energy
Qwen3-4B W4A16	QAIRT, NPU	1.7 s	15.8 W	27 J
Qwen3-4B Q4_0	llama.cpp, CPU	30.7 s	17.2 W	529 J
Qwen3.5-0.8B Q4_0	llama.cpp (Hexagon), NPU	1.6 s	19.3 W	31 J
Qwen3.5-0.8B Q4_0	llama.cpp, CPU	5.1 s	17.4 W	89 J



- **4B decode: same speed, two-thirds of the energy.** Both processors hit the same memory-bandwidth ceiling at ~18 tok/s, but the CPU got there at 21 W and the NPU at 14 W. Decode speed is set by memory; decode energy still depends on who does the work.
- **4B prefill: 20 times less energy on the NPU.** A full request, with the long prompt and a 256-token answer, comes to about 230 J on the NPU against 820 J on the CPU, 3.6 times less.
- **On the CPU, decode drew more power than prefill** (21 W against 17 W), although prefill is the compute-bound phase. Keeping the LPDDR5 busy costs power too.
- **On the 0.8B model, the CPU wins decode on both counts:** twice as fast and half the energy per token. llama.cpp's Hexagon backend drew almost as much as the CPU while generating half as fast. For prefill, even this backend saved energy: 31 J against 89 J.
- **The runtime matters for energy as much as for speed.** On the NPU, QAIRT ran the 4B model at 14 W; llama.cpp's Hexagon backend needed 19 W for a model five times smaller.

Versus the Jetson Orin Nano Super and the Raspberry Pi 5

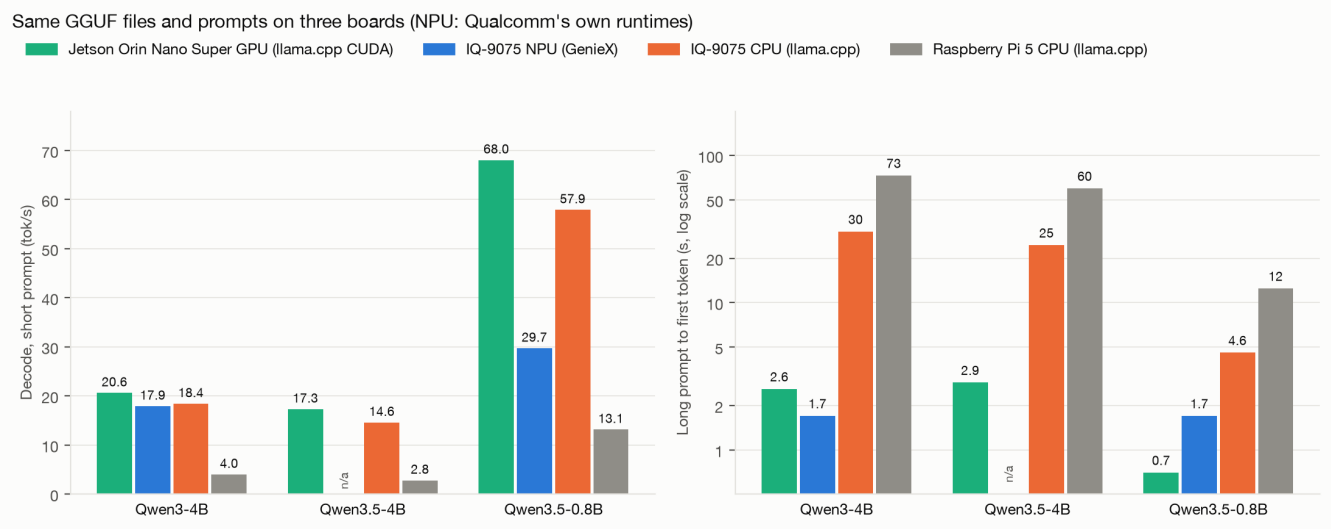
The Jetson Orin Nano Super is the closest NVIDIA board in the "Where it fits" table, and the Raspberry Pi 5 is the general-purpose reference, so I ran the LLM tests on both with the **same GGUF files** (matching SHA-256), the same prompts, seed 42, 256 tokens, and thinking disabled. All three boards ran the same llama.cpp commit (`338fce6`), built from source:

- **Jetson:** JetPack 6.2.1 (L4T 36.4.7) in MAXN_SUPER mode, CUDA build, every layer on the GPU.
- **IQ-9075:** the factory Qualcomm Linux 1.6 image, as in every other test. Because that image has no compiler, llama.cpp was built and run inside an `ubuntu:24.04` Docker container: Ubuntu supplied only the compiler and libraries, while the kernel, drivers, and hardware were Qualcomm Linux's. CPU build, 8 threads. Its speeds matched the GenieX llama.cpp runs within 3% on decode; prefill was up to 10% faster.
- **Raspberry Pi 5 (8 GB):** Raspberry Pi OS based on Debian 13, CPU build, 4 threads, official active cooler.

The new model in this round is **Qwen3.5-4B**, which has no QAIRT bundle for this chip, so on the IQ-9075 it ran on the CPU only. Each number is the mean of two runs after a warm-up; the runs agreed within 1%.

	Jetson GPU	IQ-9075 NPU	IQ-9075 CPU	Raspberry Pi 5 CPU
Qwen3-4B decode, short prompt	**20.6 tok/s**	17.9 tok/s (QAIRT)	18.4 tok/s	4.0 tok/s
Qwen3-4B, long prompt to first token	2.6 s	**1.7 s** (QAIRT)	30.4 s	73.1 s
Qwen3-4B decode, long prompt	**19.4 tok/s**	16.1 tok/s (QAIRT)	14.1 tok/s	2.0 tok/s
Qwen3.5-4B decode, short prompt	**17.3 tok/s**	—	14.6 tok/s	2.8 tok/s
Qwen3.5-4B, long prompt to first token	**2.9 s**	—	24.6 s	59.8 s
Qwen3.5-0.8B decode, short prompt	**68.0 tok/s**	29.7 tok/s (Hexagon)	57.9 tok/s	13.1 tok/s
Qwen3.5-0.8B, long prompt to first token	**0.70 s**	1.7 s (Hexagon)	4.6 s	12.5 s

The NPU column comes from the LLM section, with GenieX. For the 4B model, it is a different file: Qualcomm's W4A16 bundle for QAIRT, so only the other three columns use identical bytes.



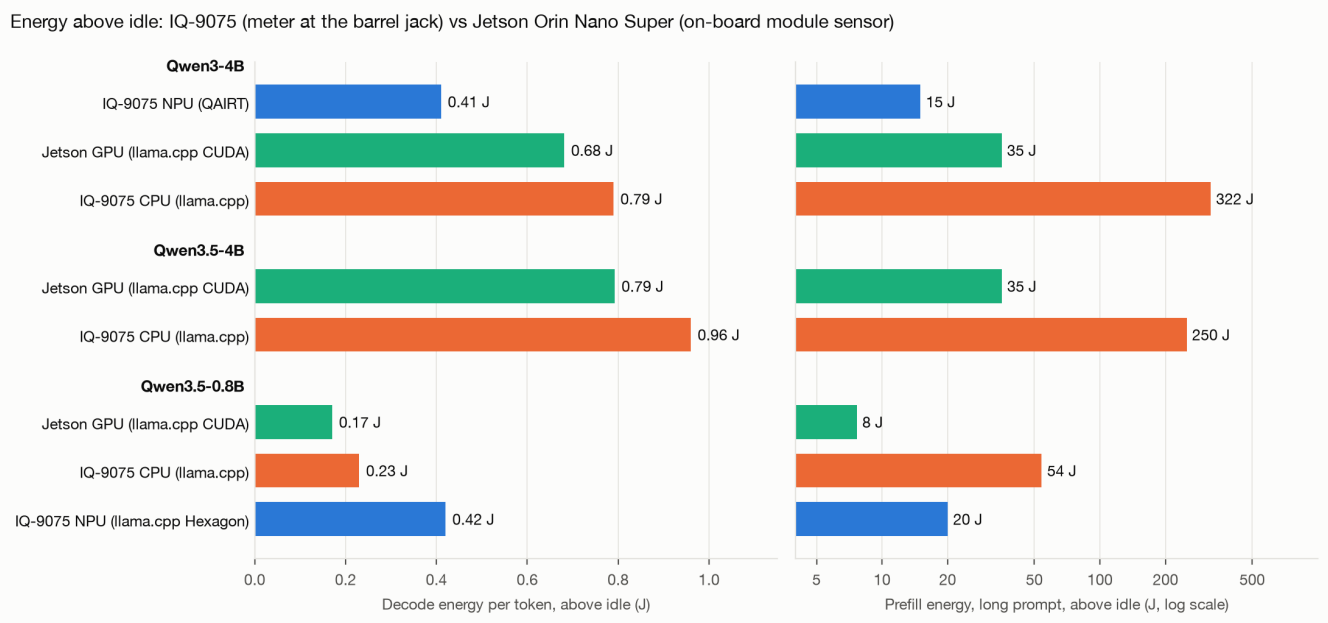
The Raspberry Pi needed a second run. The first run, with no pauses, ended at 75 °C with the throttling flag set (`vcgencmd get_throttled` = `0xe0000`). I repeated it with a pause before each request until the SoC was below 50 °C, logging temperature, clock, and throttling state every second. The board reached 77 °C with the fan at full speed, but the CPU never left 2.4 GHz and no throttling bit was active during the runs. The 4B numbers matched the first run within 3%. The 0.8B model had decoded 3 to 10% faster in the first run, apart from one outlier at 8.2 tok/s.

Energy, with a caveat. I measured the Jetson's power **without the FNB58**: its barrel jack accepts 9 to 20 V, and the 20 V trigger cable used on the IQ-9075 leaves no margin. Instead, a script read the board's own INA3221 sensor at 20 samples per second. That sensor reports the module's input rail, `VDD_IN`, so it **leaves out the carrier board (Wi-Fi card, NVMe SSD, USB, regulators) and the power supply**, and it reads lower than a meter at the barrel jack would. Its readings are also smoothed: a load step takes about 1 s to show, so decode energy comes from the steady part of the 256-token runs, and prefill energy from integrating the power over the whole long-prompt request. On the short prompt, both methods agreed within 1%. Idle on the module rail was 4.7 W, against 6.6 to 6.9 W for the whole IQ-9075 board. The Raspberry Pi's energy was not measured.

On the IQ-9075, Qwen3.5-4B was measured with the FNB58, as before, using upstream llama.cpp. In that run, Qwen3-4B on the CPU repeated the GenieX numbers of October 26 within 2% (0.79 J per token above idle, 316 J of prefill), so the two IQ-9075 sessions can be compared directly.

Because the two methods measure different things, the fairer comparison is the energy above idle, where the fixed losses of each board largely cancel:

Above idle	Jetson GPU	IQ-9075 NPU	IQ-9075 CPU
Qwen3-4B decode, per token	0.68 J	**0.41 J**	0.79 J
Qwen3-4B prefill, long prompt	35 J	**15 J**	322 J
Qwen3.5-4B decode, per token	**0.79 J**	—	0.96 J
Qwen3.5-4B prefill, long prompt	**35 J**	—	250 J
Qwen3.5-0.8B decode, per token	**0.17 J**	0.42 J	0.23 J
Qwen3.5-0.8B prefill, long prompt	**8 J**	20 J	54 J



In total energy, which flatters the Jetson here, the Qwen3-4B model used 0.91 J per token and 48 J of prefill on the Jetson, against 0.78 J and 27 J on the IQ-9075's NPU. For Qwen3.5-4B, the totals were 1.06 J and 49 J on the Jetson, against 1.43 J and 419 J on the IQ-9075's CPU, which decoded at 20.8 W.

- **The Jetson decodes faster.** Its LPDDR5 delivers 102 GB/s against 77 GB/s, and decode follows memory: 12 to 18% more tokens/s than the IQ-9075's fastest processor on every model. On Qwen3-4B, it reached about 49 GB/s effective, 48% of its theoretical bandwidth, against 55 to 57% on the IQ-9075.
- **On Qwen3-4B, the NPU is the more efficient engine.** It used 40% less energy per token above idle and less than half the prefill energy, while also finishing the prefill sooner. The sensor caveat cannot close that gap: a meter at the Jetson's barrel jack would only add to its numbers.
- **Without an NPU path, the Jetson wins.** Qwen3.5-4B ran only on the IQ-9075's CPU, which took 8.6 times longer than the Jetson's GPU to reach the first token and used 7 times more prefill energy above idle. The IQ-9075's advantage on Qwen3-4B comes from the NPU and its runtime, not from the board as a whole.
- **Qwen3.5-4B is slower than Qwen3-4B on every board**, 16 to 30% in decode. Its file is 9% larger (2.58 GB against 2.38 GB), and its hybrid architecture, with linear-attention layers, probably adds work per token. With the long prompt, it reached the first token sooner on the CPUs, but mostly because its tokenizer turned the same text into 13% fewer tokens (1,641 against 1,882). Per token, prefill ran at about the same speed on the CPUs and 21% slower on the Jetson.
- **On the 0.8B model, the Jetson wins.** It was the fastest, and it used the least energy above idle, by a margin (0.17 against 0.23 J per token) larger than the carrier board's conversion losses are likely to add. The GPU runs a small model with little overhead, while the NPU pays a fixed cost to dispatch each layer, the same lesson the IQ-9075's own CPU taught.

- **The Raspberry Pi 5 is in another class.** It decoded 4.4 to 5.2 times slower than the IQ-9075's CPU, in line with its memory: about 17 GB/s of LPDDR4X, of which Qwen3-4B used 9.5 GB/s (56%), the same fraction as the IQ-9075. Prefill was 2.4 to 2.7 times slower than on the IQ-9075's CPU, with half as many cores. More than a minute to read a 1,900-token prompt rules out RAG on a 4B model, but the 0.8B model, at 13 tok/s, is usable for short chats.
- **Peak speed and efficiency point to different boards.** For the fastest tokens and for models without a Qualcomm bundle, the Jetson; for the most work per joule on a mid-size model with a QAIRT bundle, and for prefill-heavy workloads like RAG, the IQ-9075's NPU.

Lessons and next steps

On this SoC, the scarce resource for generative AI is memory bandwidth, not TOPS. **A well-designed system puts each workload on its own block: vision on the NPU and VPU, LLM prefill on the NPU, the CPU free for logic and control**. The remaining interference comes from shared memory.

1. **An accelerator's gain depends on arithmetic intensity.** Dense CNNs gain up to 95 times; tiny or I/O-bound models gain little or lose.
2. **An accelerator is not fast at everything.** The GPU was slower than the CPU on INT8 and also used more energy, and the CPU generated text faster and more cheaply than the NPU on a 0.8B LLM.
3. **LLMs have two phases with different bottlenecks.** Prefill is compute-bound and gained 18 times on the NPU; decode is bandwidth-bound and kept the same speed, though not the same energy.
4. **The runtime matters as much as the hardware.** The same 4B model failed on llama.cpp's Hexagon backend and ran well on QAIRT, which also drew less power.
5. **Measure energy, not just speed.** The NPU sometimes drew more power than the CPU, but almost always used less energy per inference; it saved energy even on a model where it was slower.
6. **Faster memory decodes faster; more compute prefills cheaper.** The Jetson's GPU, with more bandwidth, decoded fastest, and the Raspberry Pi 5, with less than a quarter of the IQ-9075's bandwidth, decoded about 4.5 times slower. The IQ-9075's NPU did the 4B prefill faster and with less energy, but only where Qualcomm publishes a bundle for the model.
7. **The factory image is already a complete test platform.** Models, apps, Qdemo and Docker made every test possible without flashing anything.

Next steps: flash Ubuntu 24.04 Desktop, which brings `apt`, `pip`, and the documented path to AI Hub and Edge Impulse, including its ExecuTorch example that compiles PyTorch models for this board's NPU; test larger models from the Qwen3.5 and Gemma 4 families with llama.cpp, such as Qwen3.5-9B, which have no QAIRT bundle for this chip yet; test several camera streams in parallel with the multistream app; compare the HTP's power modes; and repeat the LLM tests with an 8B model, such as Qwen3-8B, which has a QAIRT bundle for the QCS9075, to find the practical limit of the 36 GB; and repeat the Jetson energy runs with a meter at its barrel jack, using a 15 V PD trigger.

Appendix: reproducible commands

All commands run as root, over SSH, on the factory Qualcomm Linux 1.6 image, except section 6, which also covers the Jetson and the Raspberry Pi 5.

1. Diagnostics

```
uname -a; nproc; free -h; df -h /
gst-inspect-1.0 | grep -iE "qtiml"
```

```
ls /usr/lib | grep -iE "qnn|snpe"
ls /etc/models /etc/configs
for z in /sys/class/thermal/thermal_zone*; do echo "$(cat $z/type): $(cat $z/temp)"; done
```

2. CPU vs NPU benchmark of every model

```
cat > /root/bench.sh <<'EOF'
#!/bin/bash
SKEL=/usr/lib/rfsa/adsp
export ADSP_LIBRARY_PATH=$SKEL
NPU="backend_type:htp;library_path:/usr/lib/libQnnHtp.so;skel_library_dir:
        $SKEL;htp_performance_mode:2"
avg() { grep "INFO: count=" | tail -1 | sed 's/.*avg=\([0-9.]*\).*/\1/'; }
printf "%-42s %9s %9s %8s\n" MODEL CPU_ms NPU_ms SPEEDUP
for m in /etc/models/*.tflite; do
  c=$(benchmark_model --graph=$m --num_threads=8 --num_runs=30 2>&1 | avg)
  n=$(benchmark_model --graph=$m --external_delegate_path=/usr/lib/libQnnTFLiteDelegate.so \
      --external_delegate_options="$NPU" --num_runs=100 2>&1 | avg)
  awk -v m="$(basename $m)" -v c="${c:-0}" -v n="${n:-0}" \
    'BEGIN{printf "%-42s %9.2f %9.3f %7.1fx\n", m, c/1000, n/1000, (n>0?c/n:0)}'
done
EOF
chmod +x /root/bench.sh && /root/bench.sh
```

For the GPU, replace the delegate options with `--use_gpu=true`.

3. Vision demos

```
export XDG_RUNTIME_DIR=/dev/socket/weston
export WAYLAND_DISPLAY=wayland-1
gst-ai-object-detection --config-file=/etc/configs/config_detection.json
gst-ai-pose-detection   --config-file=/etc/configs/config_pose.json
gst-ai-segmentation     --config-file=/etc/configs/config_segmentation.json
```

For the USB camera, copy `/etc/configs/config-usb-camera-app.json`, adjust resolution and format (`mjpeg` at 720p or 1080p) and run `gst-ai-usb-camera-app --config-file=<copy>`. The formats the camera supports show up in `gst-device-monitor-1.0 Video/Source`.

4. LLMs with GenieX

The measurements above used a long prompt in Portuguese; the English sentence below gives a comparable load.

```
alias gx="docker run -it --rm --privileged -e GENIEX_DATADIR=/data -v /root/geniex-data:/data -v /usr/
        lib:/opt/qcom-lib:ro docker.io/qualcomm/geniex:latest"
gx
# inside the container:
geniex pull unsloth/Qwen3-4B-GGUF:Q4_0
geniex pull ai-hub-models/Qwen3-4B
for i in $(seq 1 60); do echo "TinyML is the field that brings machine learning to low-power
        microcontrollers, enabling local inference on sensors."; done > /data/long.txt
echo "Summarize the text above in one sentence." >> /data/long.txt
geniex infer unsloth/Qwen3-4B-GGUF:Q4_0 -c cpu --think=false --max-tokens 64 -i /data/long.txt < /dev/
        null 2>&1 | tail -2
geniex infer ai-hub-models/Qwen3-4B --think=false --max-tokens 64 -i /data/long.txt < /dev/null 2>&1
        | tail -2
```

The container prompt (`root@<id>`) and the board prompt (`root@qcs9075-iq-9075-evk`) are easy to mix up: `docker` exists only on the board, `geniex` only in the container. In interactive chat, `/exit` leaves GenieX; `exit` leaves the container.

5. Measuring energy

Connect in this order: USB-C PD charger (unplugged from the wall) → FNB58 → 20 V PD trigger cable → barrel jack. Plug the charger into the wall, check that the FNB58 shows about 20 V and that the board boots, and only then connect the FNB58's data port to the computer.

The logger reads the FNB58 at 100 samples per second through `hidapi` and writes a CSV with time, voltage, current, power, and accumulated energy. It follows the protocol decoded by the fnirsi-usb-power-data-logger project. A second script runs each workload over SSH, writes start and end markers, and computes power and energy for each phase. Claude Opus 5.5 (Anthropic) generated both scripts under the author's direction.

```
python3 -m venv .venv && .venv/bin/pip install hidapi
.venv/bin/python fnb58_log.py -o session.csv              # one long session; Ctrl+C at the end
python3 power_bench.py run plan_models.txt -m markers.csv  # in another terminal
python3 power_bench.py report session.csv markers.csv
```

Each plan line is `label | seconds | remote command`, and an empty command is an idle phase. For example:

```
idle-00 | 30 |
inception-npu | 30 | export ADSP_LIBRARY_PATH=/usr/lib/rfsa/adsp; benchmark_model --graph=/etc/models/
inception_v3_quantized.tflite --external_delegate_path=/usr/lib/libQnnTFLiteDelegate.so --
external_delegate_options='backend_type:htp;library_path:/usr/lib/libQnnHtp.so;skel_library_dir:/usr/
lib/rfsa/adsp;htp_performance_mode:2' --num_runs=1 --min_secs={secs} --max_secs={secs} 2>&1
idle-01 | 20 |
```

For LLMs, the remote command runs `geniex infer` inside the container followed by `echo GEN_END=$(date +%s.%N)`, so the end of generation is stamped on the board's clock. In the current GenieX image, the AI Hub model appears as `qualcomm/Qwen3-4B`.

6. Upstream llama.cpp: Jetson, IQ-9075 CPU, and Raspberry Pi 5

llama.cpp built on the Jetson with CUDA (Orin's compute capability is 8.7):

```
git clone --depth 1 https://github.com/ggml-org/llama.cpp && cd llama.cpp
export PATH=/usr/local/cuda/bin:$PATH
cmake -B build -DGGML_CUDA=ON -DCMAKE_CUDA_ARCHITECTURES=87 -DLLAMA_CURL=OFF -DCMAKE_BUILD_TYPE=Relea
      se
cmake --build build -j6 --target llama-cli llama-bench llama-server
```

The prompts and the energy runs use `llama-server` through two scripts in `tools/jetson`:

```
python3 llm_prompts.py --bin llama.cpp/build/bin --models models --long prompts/longo.txt
python3 llm_energy.py --bin llama.cpp/build/bin --models models --long prompts/longo.txt -o energy.csv
```

The Jetson's CPU and GPU share its 8 GB, and the CUDA allocator does not reclaim the Linux page cache. After copying large files, the 4B model failed to load (`cudaMalloc failed: out of memory`) until the cache was dropped:

```
sudo sh -c 'sync; echo 3 > /proc/sys/vm/drop_caches'
```

For Qwen3.5-4B, `llm_energy.py --load-mode dio` loads the model with direct I/O, so it does not fill the page cache again.

The factory image on the IQ-9075 has no compiler, so llama.cpp was built inside a container that shares the host's network and folders:

```
docker run -d --name llamacpp --network host -v /root/llama:/work -v /root/geniex-data:/data
        ubuntu:24.04 sleep infinity
docker exec -it llamacpp bash
# inside the container:
apt update && apt install -y build-essential cmake git python3
cd /work && git clone https://github.com/ggml-org/llama.cpp && cd llama.cpp && git checkout 338fce6
cmake -B build -DLLAMA_CURL=OFF -DCMAKE_BUILD_TYPE=Release
cmake --build build -j8 --target llama-server llama-bench
```

The energy run pairs a script on the board, which records when each phase happened, with the FNB58 log on the Mac. `--no-stream` matters on a CPU: streaming each token over HTTP competes with the 8 inference threads and slowed decode by 5 to 20%.

```
# on the IQ-9075, inside the container, while fnb58_log.py records on the Mac
python3 llm_windows.py --bin llama.cpp/build/bin --models models --long /data/longo.txt --ngl 0 --
        threads 8 \
    --only Qwen3.5-4B-Q4_0 --no-stream -o windows.csv
# on the Mac
python3 tools/fnb58/llm_fnb58.py data/power/session-<date>.csv windows.csv -o energy.csv
```

On the Raspberry Pi 5, the same CPU build runs natively. `raspi_run_cool.sh` waits for the SoC to cool below 50 °C before each request (`llm_prompts.py --cool-below 50`) and logs temperature, clock, and `vcgencmd get_throttled` every second:

```
cmake -B build -DLLAMA_CURL=OFF -DCMAKE_BUILD_TYPE=Release && cmake --build build -j4 --target llama-
        server
./raspi_run_cool.sh
```

7. Switching OS

Flashing Ubuntu erases the factory image. The simplest path is Qualcomm Launcher (GUI, Windows, Linux or macOS). From the command line, with QDL: turn on DIP switch SW2-3 to enter EDL mode, connect USB-C to the USB0 port and flash, in this order, UFS provisioning, CDT, SAIL firmware and the Ubuntu image, following the official guide listed in the sources. The SW2 switch and the USB0 port are labeled in the board image in “The kit” section.

Sources

- Qualcomm Dragonwing IQ-9075, product page
- Qualcomm IQ-9075 Evaluation Kit (EVK)
- Arduino VENTUNO Q, store page
- Dragonwing IQ-9075 EVK, device overview
- Flash Ubuntu using QDL (IQ-9075 EVK)
- Flash Ubuntu using Qualcomm Launcher
- Day 0 on Dragonwing: verify the setup, then run your first model
- GenieX, Linux installation
- Edge Impulse, Qualcomm Dragonwing IQ-9075 EVK
- RidgeRun, Why the Qualcomm Dragonwing IQ9
- Peridio, IQ-9075 specifications
- fnirsi-usb-power-data-logger (FNB58 protocol)
- Qualcomm AI Hub models on Hugging Face

- NVIDIA Jetson Orin Nano Super Developer Kit
- llama.cpp
- Edge Impulse, PyTorch to ExecuTorch with QNN on the IQ-9075

All measurements were taken by the author. The IQ-9075 ran the factory Qualcomm Linux 1.6-ver.1.2.1 image, with the `latest` GenieX image from Docker Hub as of each date: performance on October 14, 2026, and energy on October 26, 2026. The Jetson Orin Nano Super comparison and the IQ-9075's upstream llama.cpp runs took place on October 27, 2026, and the Raspberry Pi 5 runs on October 28, 2026.

About this repository

- `figures/` : photos, charts, and diagrams used in this tutorial.
- `tools/fnb58/` : the FNB58 logger (`fnb58_log.py`), the benchmark runner and report (`power_bench.py`), the slicer (`fnb58_slice.py`), the plans run on October 26, and the scripts that draw the charts and diagrams. Set `IQ9075_HOST` and `IQ9075_KEY` to point `power_bench.py` at your board.
- `tools/jetson/` : the LLM prompt and energy scripts used on the Jetson Orin Nano Super, also used on the IQ-9075 and the Raspberry Pi 5 (`llm_prompts.py` , `raspi_run_cool.sh`). `tools/fnb58/llm_windows.py` and `llm_fnb58.py` measure llama.cpp energy on the IQ-9075 with the FNB58.
- `tools/make_board_figure.py` : the script that labeled the board image. Qualcomm's base image is not included in this repository.
- `tools/pdf/` : builds `iq-9075-tutorial.pdf` from this README with pandoc and WeasyPrint.
- `data/power/` : the raw FNB58 logs (100 samples per second), the phase markers, and the full results in `results.md` .
- `data/jetson/` : the Jetson results and how they were measured.
- `data/llamacpp/` : upstream llama.cpp speeds on the IQ-9075 CPU and the Raspberry Pi 5, with the Pi's thermal log.

Python dependencies are in `tools/requirements.txt` . The diagrams also need `rsvg-convert` (librsvg), and the PDF needs `pandoc` and `weasyprint` .

Claude Opus 5.5 (Anthropic) generated the scripts in `tools/` , the charts, and the diagrams under the author's direction. The board images are from Qualcomm and are used with permission. The FNB58 protocol comes from fnirsi-usb-power-data-logger (MIT, see `tools/fnb58/LICENSE`).